

FINAL SIX-MEMBER DEVELOPMENT WORKSTREAM SPECIFICATION

Project ID: SIH26034

Product: NyayaDrishti-LM

Sprint Window: 07 September 2026 to 13 September 2026 (6 Days)

Status: FROZEN & ASSIGNED

Workstream Overview

6-MEMBER PARALLEL WORKSTREAM ALLOCATION		
MEMBER 1: CV, OPTICS & SCALE SEMANTIC EXTRACTION • Optical Quality Gate Normalizer • ArUco & Card Homography Proximity Graph • PDP Surface Area Calculator Address NER	MEMBER 2: MULTILINGUAL OCR • DBNet++ Text Detection • PP-OCrv4 Recognizer • Tesseract Fallback	MEMBER 3: • Regex Token • Spatial • Indian Postal
MEMBER 4: STATUTORY ENGINE & INTEGRATION • Temporal Epoch Dispatcher Capture HUD • Declarative AST Rules Review Canvas • 4-State Epistemic Triage Demo Packaging	MEMBER 5: EVIDENCE & DOSSIER • SHA-256 Merkle DAG • Section 63 BSA Certificate • ReportLab PDF Generator	MEMBER 6: UI, HUD • Viewfinder & • Side-by-Side • SQLite DB &

Detailed Member Responsibilities

MEMBER 1: Lead Computer Vision & Metrology Engineer

- **Role:** Optical Quality, Calibration & Geometric Measurement
- **Primary Objective:** Deliver sub-millimeter physical scale calibration and optical gating on commodity cameras.
- **Core Tasks:**
 1. Implement Laplacian variance blur estimator ($\sigma^2 < 60 \implies \text{RETAKE}$).
 2. Implement HSV saturation specular glare detector ($V > 245, S < 15$).
 3. Implement ArUco fiducial marker detector and ISO 7810 card contour detector.
 4. Compute 3×3 planar homography matrix H via OpenCV Direct Linear Transform.
 5. Warp perspective to orthogonal plane and compute metric scale factor $S = \text{mm/pixel}$.
 6. Calculate Principal Display Panel (PDP) surface area in cm^2 for rectangular and cylindrical packaging.
- **Deliverables:** `core/quality_gate/` and `core/homography/` Python modules with unit tests.
- **Interface Output:** `QualityGateResult` and `CalibrationResult` JSON schemas.
- **Acceptance Criteria:** $\text{MAE} \leq 0.15 \text{ mm}$ against vernier caliper measurements on planar test targets.

MEMBER 2: Deep Learning & Multilingual OCR Engineer

- **Role:** Perception Engine & Scene Text Recognition
- **Primary Objective:** High-speed multilingual text extraction across English and Devanagari Hindi on CPU.
- **Core Tasks:**
 1. Export DBNet++ text detector to ONNX and apply INT8 Post-Training Quantization.
 2. Export PaddleOCR PP-OCRv4 (SVTR) recognizer to ONNX INT8 with Latin and Devanagari dictionaries.
 3. Implement polygonal bounding box crop extraction and coordinate normalizer.
 4. Implement secondary consensus pass with Tesseract v5 on low-confidence crops ($\text{conf} < 0.65$).
 5. Measure character-level confidence scores and construct token list.
- **Deliverables:** `core/ocr_engine/` ONNX pipeline with benchmark test script.
- **Interface Output:** `OCROutput` JSON schema with polygons, text strings, and confidences.

- **Acceptance Criteria:** CER $\leq 3.0\%$ on printed text; total inference latency ≤ 200 ms on CPU per panel.

MEMBER 3: Information Extraction & NLP Engineer

- **Role:** Semantic Token Linking & Entity Normalization
- **Primary Objective:** Convert raw OCR tokens into verified statutory packaging entities without hallucinations.
- **Core Tasks:**
 1. Implement deterministic regex parsers for MRP, tax clauses, Net Quantity, and Unit Sale Price.
 2. Implement prohibited metric unit symbol detector (flagging gms, gm, Kgs, ML, ltrs).
 3. Build 2D spatial proximity graph (K-D Tree) to link key-value pairs on packaging labels.
 4. Implement Devanagari numeral converter mapping Indic digits (०..९) to IEEE floating-point numbers.
 5. Implement Indian postal address segmenter extracting State and 6-digit PIN codes.
 6. Implement Consumer Care 4-tuple completeness verifier (Contact, Address, Phone regex, Email regex).
- **Deliverables:** core/extractor/ Python module with comprehensive regex test suite.
- **Interface Output:** NormalizedCommodityFacts JSON schema.
- **Acceptance Criteria:** Exact Field Match $\geq 95\%$ on synthetic and physical pilot datasets.

MEMBER 4: Legal Metrology & Rule Engine Architect

- **Role:** Statutory Compliance Logic & Temporal Reasoning
- **Primary Objective:** Translate the Legal Metrology Act and LMPC Rules into an auditable, deterministic AST engine.
- **Core Tasks:**
 1. Create immutable JSON rule snapshots for each statutory milestone (2011 Base, 2017 Font, 2021 USP, 2023 Jan Vishwas).
 2. Implement Temporal Statutory Epoch Dispatcher matching product Mfg Date to rule snapshot.
 3. Implement Table-I font-height evaluator comparing measured mm against required threshold by PDP area.

4. Implement Unit Sale Price (USP) arithmetic consistency checker: $\|(\text{USP} \times \text{NetQty}) - \text{MRP}\| \leq 0.02$.
 5. Implement Net Quantity exclusion space margin checker (clearance $\geq 1 \times V, 2 \times H$).
 6. Implement 4-State Epistemic Verdict Triage (**PASS**, **FAIL**, **REVIEW**, **UNABLE_TO_VERIFY**).
- **Deliverables:** **core/rule_engine/** AST evaluator, rule definition JSONs, and statutory test suites.
 - **Interface Output:** **ComplianceVerdictResult** JSON schema with exact Gazette GSR citations.
 - **Acceptance Criteria:** 100% rule traceability; zero hallucinations; execution time < 5 ms.

MEMBER 5: Security, Cryptography & Evidentiary Dossier Engineer

- **Role:** Cryptographic Chain of Custody & Statutory PDF Generator
- **Primary Objective:** Ensure court admissibility under Section 63 BSA 2023 and generate formal inspection memos.
- **Core Tasks:**
 1. Implement SHA-256 Merkle DAG chaining raw images, calibration matrices, OCR tokens, and rule findings.
 2. Implement device fingerprinting, GPS coordinate binding, and network atomic time stamping.
 3. Implement Section 63 BSA electronic certificate generator with officer digital signature block.
 4. Implement ReportLab PDF generator producing formatted, tamper-evident statutory Inspection Memos (Form 1).
 5. Embed visual crop overlays, measured millimeter deficits, and verification QR code in PDF memo.
 6. Create mock eMaap integration endpoint exporting standardized JSON payloads.
- **Deliverables:** **core/evidence/** module, PDF generation engine, and Merkle test suite.
- **Interface Output:** **EvidenceObject** JSON schema and ready-to-print **InspectionDossier.pdf**.
- **Acceptance Criteria:** Validated Merkle tree root; signed PDF renders in < 1.5 seconds.

MEMBER 6: Lead Frontend, UX & Integration Engineer

- **Role:** Cross-Platform UI, Viewfinder HUD & End-to-End System Packaging
- **Primary Objective:** Intuitive field officer user experience running 100% offline with zero lag.
- **Core Tasks:**
 1. Build React + Vite cross-platform UI with Electron desktop wrapper and responsive tablet layout.
 2. Implement live camera viewfinder with real-time HUD overlays for blur and specular glare feedback.
 3. Implement guided multi-panel capture workflow tracking progress across package facets (Front, Back, Sides).
 4. Implement Side-by-Side Review Canvas displaying original photo, rectified crop, millimeter grid, and rule citations.
 5. Implement local encrypted SQLite storage via offline storage adapter and search registry.
 6. Package the complete application, models, and demo dataset into a one-click runnable distribution.
- **Deliverables:** `client/` frontend application, Electron container, and demo launch script.
- **Interface Output:** Responsive graphical user interface interacting with local Python/FastAPI backend.
- **Acceptance Criteria:** < 100 ms UI responsiveness; zero crashes during continuous 10-inspection stress test.